

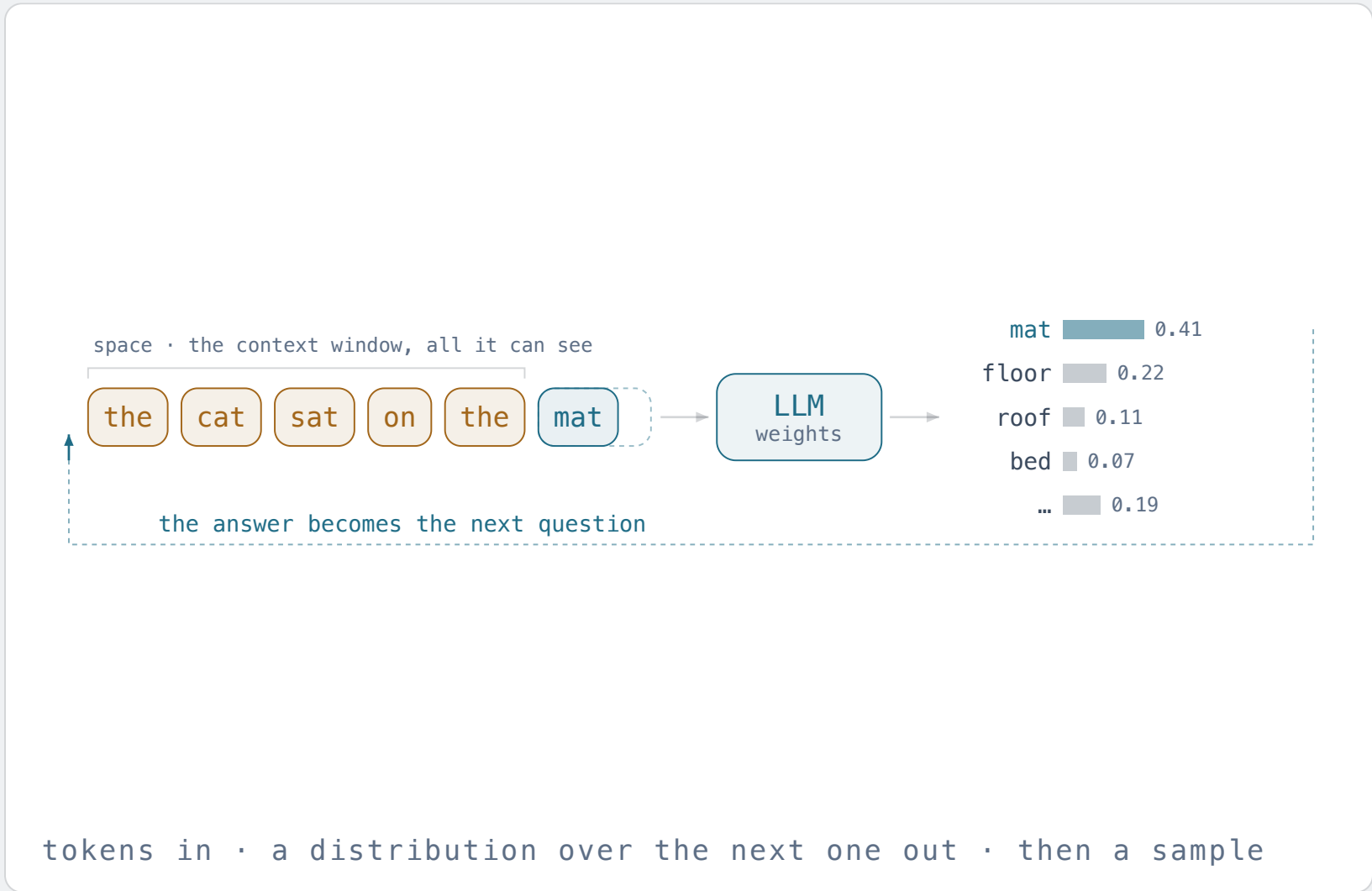
Generated Code

What a large language model is; a generated program as notation;
the compiler, the test and the model checker as the gate.

← CC 12 · Bounded Model Checking

WHAT IT IS

A large language model is a machine trained on **notation**, asked for **meaning**.



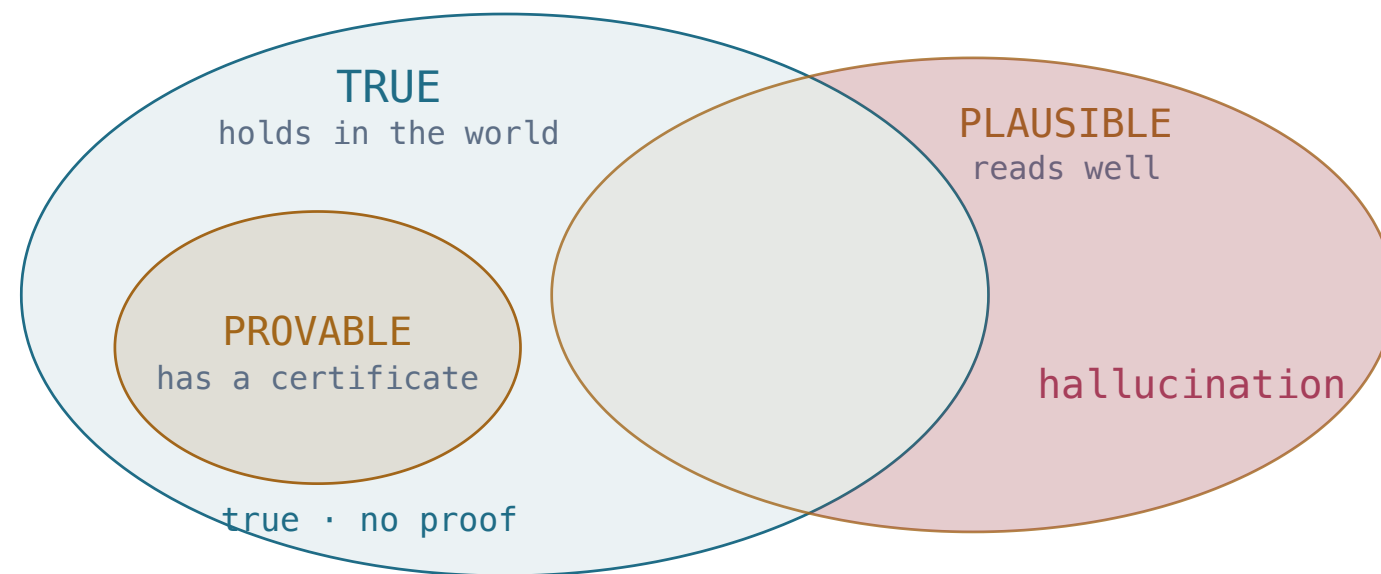
Context in, a probability for every next token out, one sampled and appended. Trained by predicting the next token: a purely syntactic signal.

It is arithmetic on bits, in principle a C* program, which mipster would run. It computes what a Turing machine computes, and week 8 said what that means for self-certification.

Its behaviour on every context is one of the introduction class's infinite answer sheets: only sampled, never enumerated.

HALLUCINATION

A hallucination is a **proof-shaped object** that isn't **true**.



plausible · provable · true

Fluency is syntax. Correctness is semantics. A generated program can be perfectly fluent C — indented, idiomatic, commented — and not parse as C*, or parse and divide by zero, or run and compute the wrong thing. Three tests, and they are not the same test.

Not a defect to patch away: the gap between notation and meaning, at consumer scale. The response is the same as everywhere in this class: check the notation against a semantics you own.

Ask a machine for a C* program. Then *starc*, *mipster*, *rotor*.

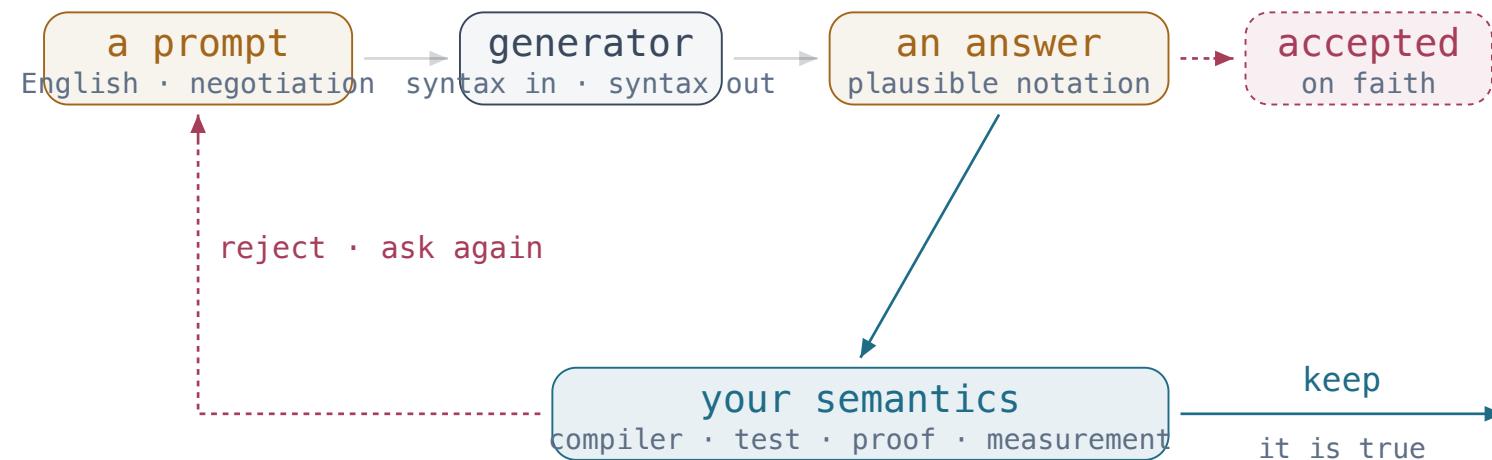
```
$ ./selfie -c generated.c
./selfie: syntax error in generated.c in line 3:
  unexpected symbol "[" found
// ... after asking again for C*, not C:
$ ./selfie -c generated.c -m 1 7
5040
$ ./rotor -c generated.c - 0
$ tools/bitme.py -kmax 1000 --use-bitwuzla \
    generated-rotorized.btor2
... core-0-division-by-zero - satisfiable · input 0
```

Does it parse? The compiler decides, always.
Usually not on the first try: the machine
produces C fluently, and C* is not C.

Does it do what was asked? Run it on the inputs
you thought of. Presence, not absence.

Can it fail? Within a thousand steps, on any
input: rotor and bitme. Absence, within the
bound. None of the three is available to the
machine that wrote the program, because the
only semantics in the loop is the one you
brought.

A generator supplies notation. The semantics has to come from **somewhere else**.



the only part the machine does not supply

what the machine supplies · and what it does not

Cantor's list needed an object built from outside it. Gödel's system needed a stronger one. Thompson's compiler needed a second compiler. The generator needs a check it cannot be. One theorem, four costumes, and this class met all four.

Models grading models, models trained on model output, agents invoking themselves: the loops are closed, and week 8 said what no loop establishes about itself. Spend the effort on the outside check.

WHAT CHANGED

Generation got **cheap**. Verification did **not**. This class was about verification.

Week 11: hard to find, easy to check. That is a description of a healthy relationship with a machine: accept work you could not have produced, provided you can check it. When you cannot, delegation is faith.

Value migrates to the two ends the machine does not occupy: stating what should be true, and establishing that it is. A grammar, a type, a test, a bad state and a bound. Every one of those is something you wrote this semester.

FOR THE CAPSTONE

Use a machine to draft struct-execution if you like. Then run the three checks, and note in your submission which of them found what. That note is worth more than the draft.

Listen, then read.

LISTEN

[Strauss · Capriccio](#) — Runnicles, San Francisco, Kiri Te Kanawa. His last opera argues for an evening about which comes first, the words or the music: notation or meaning. The Countess never chooses, and the gate is the reason she cannot.

READ

[The Alignment Problem](#) — Christian, 2020: what it means to get a machine to do what was wanted, told through the people who tried. And, more technical, [Evaluating Large Language Models Trained on Code](#) — Chen and others, 2021: the Codex paper, with the honest table of how often the generated program passes the tests.